

Project 3 – Secure Linux Server Setup & Hardening

Project Report

Vulnerability Assessment and Security Validation
of a Hardened Ubuntu Linux Server Using
Greenbone Vulnerability Manager (OpenVAS)

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1. Executive Summary

This project focused on assessing the security posture of a hardened Ubuntu Linux server within a virtual lab. Ubuntu Server was hardened using industry best practices, followed by network enumeration using Nmap and preparation of Greenbone Vulnerability Manager (OpenVAS) for vulnerability assessment. The OpenVAS environment was successfully installed and configured. At the submission deadline, the remaining Greenbone feeds (SCAP, CERT, and GVM Data) were still synchronizing, preventing execution of the final vulnerability scan.

2. Introduction

The objective of this project was to demonstrate enterprise-style Linux server hardening and vulnerability assessment using industry-standard cybersecurity tools.

3. Objectives

1. Deploy and harden Ubuntu Server.
2. Configure secure SSH access.
3. Enable firewall, auditing and intrusion prevention.
4. Perform host discovery and service enumeration using Nmap.
5. Configure Greenbone/OpenVAS.
6. Prepare automated vulnerability assessment.

4. Lab Environment

Component	Details
Host Machine	Apple MacBook Air M2
Virtualization	UTM
Attacker Machine	Kali Linux
Target Machine	Ubuntu Server 26.04 LTS
Vulnerable Machine	Metasploitable 2

Network	Bridged
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5. Network Architecture

1. Kali Linux: 192.168.1.8
2. Ubuntu Server: 192.168.1.6
3. Metasploitable 2: 192.168.1.7

6. Security Hardening Overview

1. System updated
2. SSH configured
3. Root login disabled
4. UFW enabled
5. Fail2Ban installed
6. Auditd configured
7. Lynis audit completed

7. Assessment Methodology

Host Discovery → Port Scan → Service Detection → OS Detection → Vulnerability Enumeration → OpenVAS Configuration → Feed Synchronization → Target Configuration.

8. Network Enumeration using Nmap

1. Host Discovery
`"nmap -sn 192.168.1.0/24"`
2. Port Scan
`"nmap 192.168.1.6"`
3. Version Detection
`"nmap -sV 192.168.1.6"`
4. OS Detection
`"sudo nmap -O 192.168.1.6"`

5. Aggressive Scan

```
"sudo nmap -A 192.168.1.6"
```

6. Vulnerability Scan

```
"sudo nmap --script vuln 192.168.1.6"
```

9. OpenVAS (Greenbone) Configuration

Greenbone Vulnerability Manager, GSAD, OSPD-OpenVAS, PostgreSQL and Redis were configured successfully. Target and scan configuration were completed.

10. Feed Synchronization Status

Feed	Status
NVT	Current
SCAP	Update in Progress
CERT	Update in Progress
GVMD Data	Update in Progress

The remaining feeds were still synchronizing at the time of submission, preventing execution of the vulnerability scan.

11. Target & Scan Configuration

7. Target: Ubuntu Server

8. Host: 192.168.1.6

9. Scan Profile: Full and Fast

10. Scanner: OpenVAS Default

12. Observations

Ubuntu Server hardening, Nmap enumeration, OpenVAS installation, feed synchronization, target creation and scan configuration were completed

successfully. Final automated vulnerability scanning remained pending due to incomplete feed synchronization.

13. Security Recommendations

Complete feed synchronization.
Run Full and Fast vulnerability scan.
Apply remediation based on findings.
Schedule regular scans.
Keep the operating system updated.

14. Challenges Encountered

Storage limitations required reinstallation of the virtual environment.
Extended Greenbone feed synchronization delayed scanner readiness before the submission deadline.

15. Conclusion

The project successfully demonstrated Linux server hardening and preparation of an enterprise vulnerability assessment environment. Although the final OpenVAS scan could not be completed before submission because feed synchronization was still in progress, the environment was fully configured and ready for scanning once synchronization completes.

16. References

- Kali Linux Documentation
- Greenbone Community Documentation
- Ubuntu Server Documentation
- Nmap Reference Guide
- CIS Ubuntu Benchmark